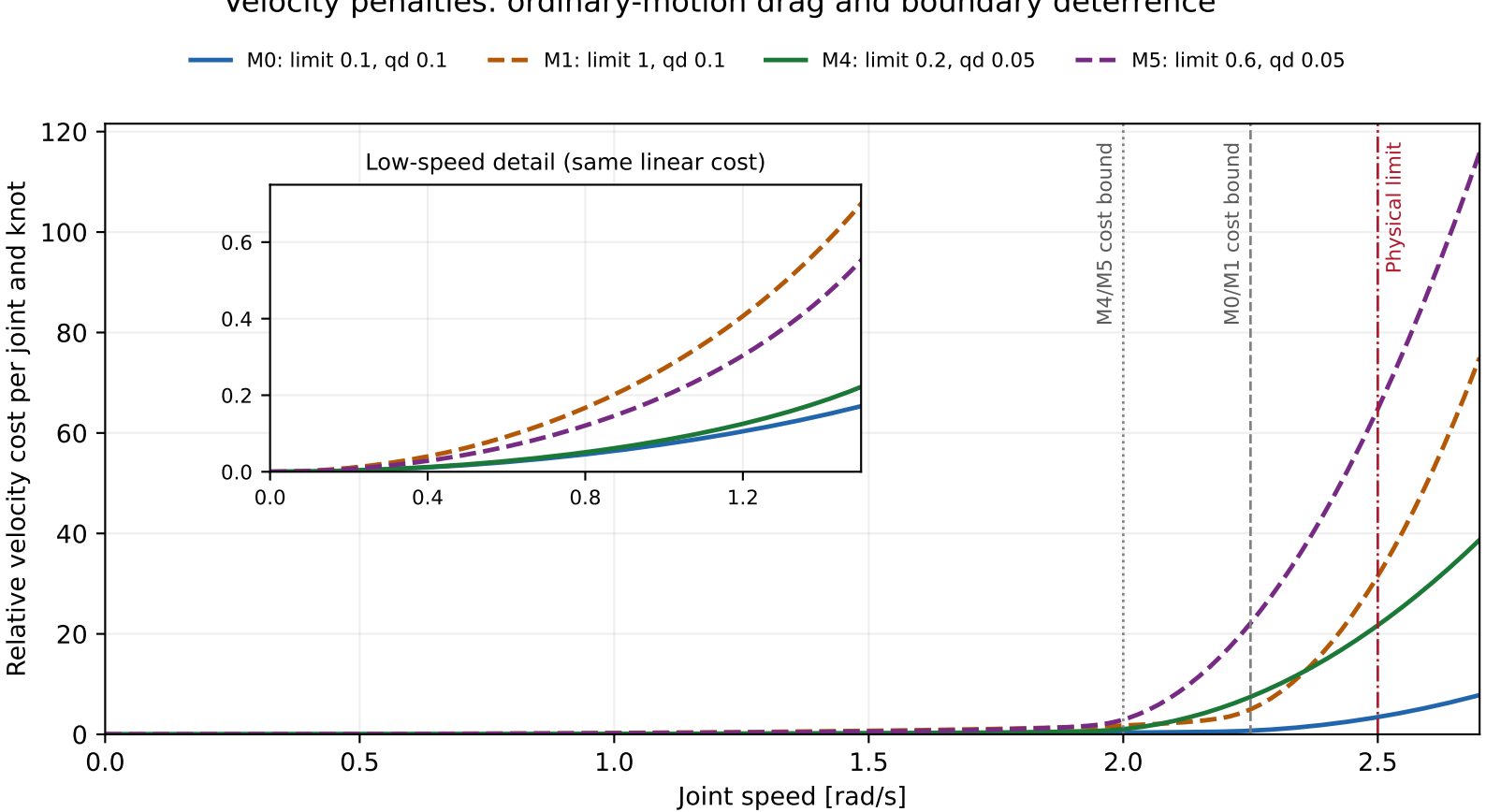
Velocity penalties: ordinary-motion drag and boundary deterrence
M0: limit 0.1, qd 0.1
M1: limit 1, qd 0.1
M4: limit 0.2, qd 0.05
M5: limit 0.6, qd 0.05
Low-speed detail (same linear cost)
M4/M5 cost bound
M0/M1 cost bound
Physical limit
Relative velocity cost per joint and knot
Joint speed [rad/s]
C2 continuation uses 1% of each cost interval. Curves omit pose/effort costs and dynamics.